Training Vectors: SVM QP Formulation

Training Vectors:

x1 = (1, 1), y1 = 1
x2 = (2, 2), y2 = 1
x3 = (7, 4), y3 = 1
x4 = (4, 8), y4 = -1

Use C = 1

Maximize $L = \sum \alpha_i - \frac{1}{2} \sum \sum \alpha_i \alpha_j y_i y_j x_i^T x_j$

$\Rightarrow$ Minimize $L_D(\alpha) = \frac{1}{2} \sum \sum \alpha_i \alpha_j y_i y_j x_i^T x_j - \sum \alpha_i$

$\alpha_1 y_1 + \alpha_2 y_2 + \alpha_3 y_3 + \alpha_4 y_4 = 0$

Minimize: [$+\alpha_1^2 + 2\alpha_1\alpha_2 - 5.5\alpha_1\alpha_3 - 6\alpha_1\alpha_4 + 2\alpha_2\alpha_1 + 4\alpha_2^2 - 11\alpha_2\alpha_3 - 12\alpha_2\alpha_4$

$- 5.5\alpha_3\alpha_1 - 11\alpha_3\alpha_2 + 32.5\alpha_3^2 + 36\alpha_3\alpha_4 - 6\alpha_4\alpha_1 - 12\alpha_4\alpha_2 + 36\alpha_4\alpha_3 + 40\alpha_4^2$]

$- [\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4]$

Subject to: $\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4 = 0$

$\alpha_1, \alpha_2, \alpha_3, \alpha_4 \leq 1$ (C)

$\alpha_1, \alpha_2, \alpha_3, \alpha_4 \geq 0$

Quadratic Equation in α with constraints